

BAIS: International Relations

Security Dilemmas and Nuclear Proliferation

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In-Class Survey

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Roadmap

Before the break

1. Review: IR Theories and Key Concepts
2. The Security Dilemma
3. Nuclear Weapons: An Overview

After the break

4. Why Do States Build the Bomb?
5. The Proliferation Debate
6. Controlling Proliferation

Note

A supplementary bibliography for all scholars referenced in this lecture is available on Brightspace.

Review: IR Theories and Key Concepts

Last Week: Three Traditions of IR

	Realism	Liberalism	Constructivism
Core Assumption	Anarchy produces competition and conflict	Cooperation is possible under anarchy	Ideas and identity shape state interests
Key Concepts	Power, self-help, balance of power, security dilemma	Institutions, interdependence, regimes, democratic peace	Norms, identity, social construction, cultures of anarchy
View of Conflict	Enduring feature of international politics	Reducible through institutions and interdependence	Depends on how states perceive each other

Key Realist and Liberal Concepts

From realism

- **Anarchy** and the absence of a world government
- **Self-help** – states must provide for their own security
- **Balance of power** – preventing any single state from dominating
- **Internal balancing** (military build-up) vs. **external balancing** (alliances)
- **Security dilemma** – efforts to increase security can threaten others

From liberalism

- **International institutions** reduce uncertainty and facilitate cooperation
- **Regimes** – rules and norms governing state behavior in specific issue areas
- **Interdependence** raises the costs of conflict
- **Arms control** as an institutional response to security competition

Key Constructivist Concepts

- **Norms** shape what states consider appropriate behavior – not just material interests
- **Identity** determines how states define their interests and roles in the world
- **Social construction** – the meaning of anarchy, power, and weapons is not fixed but created through interaction
- **Taboos and stigma** – shared beliefs can prohibit certain actions even when they might be strategically advantageous

Connecting forward

Each of these traditions offers a different lens for understanding why states build nuclear weapons – and why some choose not to.

The Security Dilemma

What Is the Security Dilemma?

Definition

When one state's efforts to increase its own security make other states less secure, triggering a spiral of competition.

Key dynamics:

- Defensive measures perceived as offensive threats
- Uncertainty about intentions in an anarchic system
- Can lead to arms races even when no state desires conflict

The security dilemma is a foundational concept for understanding why states arm – and why arms races emerge.

From Security Dilemma to Arms Race

The security dilemma logic applied to weapons:

- State A arms → State B feels threatened → State B arms → State A feels threatened → spiral
- **Arms race:** an action-reaction cycle of military build-up

Historical illustration

The U.S.-Soviet nuclear arms race – from a handful of warheads in 1945 to over 70,000 at the Cold War peak.

The “chain reaction” logic: one state’s bomb triggers neighbors to seek their own – “proliferation begets proliferation.”

Nuclear Weapons: An Overview

The Nuclear Club: Who Has the Bomb?

State	First Test	NPT Status
United States	1945	NPT nuclear-weapon state
Russia (USSR)	1949	NPT nuclear-weapon state
United Kingdom	1952	NPT nuclear-weapon state
France	1960	NPT nuclear-weapon state
China	1964	NPT nuclear-weapon state
Israel	undeclared	Not a signatory
India	1974	Not a signatory
Pakistan	1998	Not a signatory
North Korea	2006	Withdrew 2003

States that gave up weapons or programs: South Africa, Ukraine, Kazakhstan, Belarus, Libya

Nuclear Arsenals Over Time

Key trends:

- **1945:** The U.S. is the sole nuclear power – two bombs used on Hiroshima and Nagasaki
- **1945–1986:** Rapid build-up during the Cold War arms race
- **Late 1980s peak:** ~70,000 warheads worldwide (overwhelmingly U.S. and Soviet)
- **Post-Cold War:** Sharp reductions through arms control treaties
- **Today:** ~12,500 warheads globally – but modernization programs continue across all nine nuclear-armed states

Key distinction

Horizontal proliferation – the spread of nuclear weapons to *new* states.

Vertical proliferation – increasing the size and sophistication of *existing* arsenals.

Sources: Heywood & Whitham, Fig. 12.1; Arms Control Association; SIPRI

Global Nuclear Arsenals Today

State	Est. Warheads	Notes
Russia	~6,375	Largest arsenal; extensive modernization
United States	~5,800	Triad: land, sea, air delivery systems
China	~320	Rapidly expanding; may triple by 2035
France	~290	Sea- and air-based
United Kingdom	~215	Sea-based (Trident)
Pakistan	~160	Fastest-growing arsenal
India	~150	Developing triad capability
Israel	~90	Policy of deliberate ambiguity
North Korea	~40–50	Testing ongoing; delivery systems advancing

Source: Arms Control Association / SIPRI (as cited in Heywood & Whitham, Fig. 12.2)

Key Concepts in Nuclear Security

- **Nuclear deterrence** – preventing attack by threatening nuclear retaliation
- **Extended deterrence** – protecting allies under the “nuclear umbrella”
- **Mutual assured destruction (MAD)** – both sides can destroy each other, creating stability (in theory)
- **First-strike vs. second-strike capability** – ability to attack first vs. survive and retaliate
- **Nuclear taboo** – normative inhibition against using nuclear weapons
- **Stability-instability paradox** – nuclear stability at the strategic level may enable conventional conflict

Three Perspectives on Nuclear Proliferation

Approach	Core Logic	Policy Implication
Realist	States acquire nuclear weapons to survive in an anarchic system; proliferation is driven by security threats.	Balance of power, deterrence
Liberal	International institutions and treaties can constrain proliferation; cooperation is possible.	Arms control regimes (NPT, IAEA, CTBT)
Constructivist	Norms, identity, and prestige shape proliferation decisions; the “nuclear taboo” restrains use.	Norm-building, stigmatization, socialization

Adapted from Heywood & Whitham, “Approaches to...” sidebar

Break

We will resume in 10 minutes.

Survey Results

Let's see how you did – and what the experiment revealed.

Why Do States Build the Bomb?

Three Models (Overview)

Sagan (1996): “Why Do States Build Nuclear Weapons?”

The Security Model

States build nuclear weapons to increase national security against external threats.

The Domestic Politics Model

Nuclear weapons are political tools used by domestic actors – military, bureaucrats, politicians – to advance parochial interests.

The Norms Model

Nuclear weapons serve as symbols of national identity, modernity, and international status.

“Nuclear weapons... are more than tools of national security; they are political objects of considerable importance in domestic debates and internal bureaucratic struggles and can also serve as international normative symbols of modernity and identity.” (Sagan, 1996, p. 55)

The Security Model

- Rooted in realist and neorealist theory
- Core logic: states in an anarchic system respond to external threats
- “Chain reaction” – one state’s bomb triggers neighbors to seek their own
- Nuclear weapons as the ultimate deterrent and guarantor of survival

Case: The Soviet Union (1949)

After Hiroshima, Stalin told his scientists: “Provide us with atomic weapons in the shortest possible time. You know that Hiroshima has shaken the whole world. The balance has been destroyed. Provide the bomb – it will remove a great danger from us.” (Sagan, 1996, p. 58)

Also applies to: Britain, France, and China, which built weapons in response to perceived threats. “Proliferation begets proliferation” (Sagan, 1996, p. 57).

The Domestic Politics Model

Nuclear weapons serve bureaucratic and political interests *within* the state.

Three key domestic actors:

- **Nuclear energy establishment** – scientists and labs seeking funding and prestige
- **Military** – services and units that would gain resources and mission from a nuclear arsenal
- **Politicians** – leaders who use nuclear programs for domestic political advantage

Case: India (1974)

India's first nuclear test was ordered by Gandhi "with the advice of a very small circle of personal advisers and scientists from the nuclear establishment" (Sagan, 1996, p. 67). Despite China testing in 1964, India waited a *decade* – a delay the security model cannot easily explain.

The Norms Model

- “State behavior is determined not by leaders’ cold calculations about the national security interests... but rather by deeper norms and shared beliefs” (Sagan, 1996, p. 73)
- Nuclear weapons as symbols of **modernity**, technological achievement, and great-power status
- States may pursue the bomb to *join the club* – or refrain because of normative commitments
- The nuclear taboo works in both directions: stigmatizing use *and* shaping the identity of non-nuclear states

Case: France (1960)

De Gaulle told Eisenhower in 1959: “A France without world responsibility would be unworthy of herself... it is for this reason too that she intends to provide herself with an atomic armament.” (Sagan, 1996, p. 79)

Sagan's Three Models: Summary

Model	Core Logic	Key Actors	Case Example
Security	External threats in anarchy	State leaders responding to threats	USSR (1949)
Domestic Politics	Bureaucratic and political interests	Scientists, military, politicians	India (1974)
Norms	Identity, prestige, status	State identity, international norms	France (1960)

Sagan's conclusion

“The evidence... strongly suggests that **multicausality**, rather than measurement error, lies at the heart of the nuclear proliferation problem.” (Sagan, 1996, p. 85)

Beyond Sagan: Additional Explanations

Subsequent scholars have identified further factors shaping proliferation decisions:

- **Leader psychology** – personal identity and “oppositional nationalism” can drive proliferation decisions (Hymans, 2006)
- **Political economy** – domestic coalitions shape nuclear choices; outward-oriented economies are less likely to proliferate (Solingen, 2007)
- **Strategic culture** – how states think about the role of force and the meaning of nuclear weapons (Kartchner & Larsen, 2009)
- **Leverage and bargaining** – using a nuclear program as a diplomatic tool to extract concessions

Think about it

Which combination of factors best explains North Korea's nuclear program?

See Greitens (2022) for a synthesis of these additional explanations.

The Proliferation Debate

“More May Be Better” (The Optimist View)

The Waltz-Sagan debate (see also Heywood & Whitham, Ch. 12)

If nuclear deterrence works, then spreading nuclear weapons could *increase* stability.

- Rational states will avoid war when both sides face nuclear annihilation
- Evidence cited: no nuclear war since 1945
- Nuclear-armed rivals (U.S.-USSR, India-Pakistan) have avoided direct conflict
- The Cold War as “the long peace”

Core logic

Deterrence produces caution. The spread of nuclear weapons extends that caution to more states.

“More Will Be Worse” (The Pessimist View)

Organizations are not perfectly rational – they have biases, routines, and pathologies.

Key risks:

- **Preventive war** – states may attack *before* a rival completes its nuclear program
- **Accidents and unauthorized use** – near-misses during the Cold War (1983 Soviet false alarm, 1961 Goldsboro incident)
- **Civil-military dysfunction** – military organizations may prioritize offensive doctrines, undermining stable deterrence
- **Irresponsible nuclear powers** – not all states have robust command-and-control systems

The more states with nuclear weapons, the greater the probability of catastrophic failure.

The Proliferation Debate: Summary

Waltz vs. Sagan; Heywood & Whitham, Ch. 12, “Debating...” box

“More May Be Better”

- Absence of nuclear war since 1945
- Effective deterrence
- International stability
- Nuclear statesmanship

“More Will Be Worse”

- Fallibility of deterrent systems
- Danger of nuclear imbalances
- “Useable” nuclear weapons
- Irresponsible nuclear powers

The question remains

Does nuclear deterrence make the world safer – or does each new nuclear state increase the risk of catastrophe?

Controlling Proliferation

The Non-Proliferation Regime

Key institutions and treaties:

- **NPT (1968)** – three pillars: non-proliferation, disarmament, peaceful use of nuclear energy
- **IAEA** – verification, safeguards, and inspections
- **CTBT (1996)** – comprehensive ban on nuclear testing (not yet in force)
- **Nuclear-weapon-free zones** – regional agreements (Latin America, Southeast Asia, Africa, Central Asia, South Pacific)

Arms control vs. disarmament:

- **Arms control** – managing and limiting arsenals through bilateral or multilateral agreements
- **Disarmament** – eliminating entire categories of weapons

Arms Control: Key Agreements

Year	Agreement
1963	Limited Test Ban Treaty (LTBT) – banned atmospheric, underwater, and outer space tests
1968	Nuclear Non-Proliferation Treaty (NPT) – cornerstone of the non-proliferation regime
1972	SALT I / ABM Treaty – first U.S.-Soviet limits on strategic weapons
1987	Intermediate-Range Nuclear Forces Treaty (INF) – eliminated an entire class of missiles
1991	START I – major reductions in U.S. and Soviet strategic arsenals
1996	Comprehensive Test Ban Treaty (CTBT) – opened for signature; not yet in force
2010	New START – limited each side to 1,550 deployed strategic warheads
2017	Treaty on the Prohibition of Nuclear Weapons – adopted at UN; nuclear powers have not signed

Source: Adapted from Heywood & Whitham, Ch. 12

Contemporary Challenges

The unraveling of arms control:

- **2019:** U.S. withdraws from the INF Treaty (citing Russian violations)
- **2023:** Russia suspends participation in New START
- **February 5, 2026:** New START expires without renewal – the first time in over 50 years with *no binding limits* on U.S. and Russian nuclear arsenals

Other ongoing challenges:

- Iran nuclear deal (JCPOA) – collapsed after U.S. withdrawal (2018)
- North Korea's advancing program – tested ICBMs capable of reaching the continental U.S.
- China's rapid nuclear expansion
- Modernization programs across all nine nuclear-armed states

Sources: CNN, NPR, *Washington Post*, UN News (February 2026)

Wrapping Up

Key Takeaways

1. The **security dilemma** helps explain why states arm – including with nuclear weapons
2. Nine states currently possess nuclear weapons; the global arsenal peaked at ~70,000 warheads and now stands at ~12,500
3. Sagan's three models show proliferation is driven by **security, domestic politics, and norms** – not just one factor
4. The **Waltz-Sagan debate** on whether proliferation increases or decreases stability remains unresolved
5. The arms control regime that reduced arsenals after the Cold War is under serious strain

Next Week

Week 4: Non-Western Approaches to IR

Readings and recommended materials listed on Brightspace.

Thank you!

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